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"15 = The estimated number of nuclear power plants that would be needed to meet the expected increase in energy consumption by AI systems." So. We're ok with nuclear power plants again?

Posted by u/Lago795 • • 0 points • 7 comments

Here's an idea about why Bill G might be buying up all that farmland: he's going to lease it to Microsoft so they can build 15 new nuclear power plants. Because *Microsoft* is going to build their own standard-size nuclear reactors.

Here all this time I thought he was just going to do some robot enhanced, glyphosate enriched food production for the masses. After reading the article about AI's power needs, though, it made me think again.

Bill can charge MS some big $$ to lease the land, win for him. Approvals slide through the government like grease in a palm. Otherwise, where would they realistically be built? Not In My Back Yard, I hope.

Yeah, and also: since when are we collectively okay with building new nuclear power plants again? I thought the enlightened way was wind and solar! Where are Bruce Springsteen and the No Nukes Concert people?

And another thing: can nobody say "NO!" to AI? Here's what I see: 1. AI is created. 2. AI says "I need more power." 3. We just give it more power? We can't say "sorry, we can live without the cloud"? 4. Looks like people haven't been paying attention to books and movies where the AI turns out to be the evil mastermind.

But wait, there's more! How about that climate change, eh? Does anyone ever mention the heat generated from all these massive data servers and whatnot? YOU better watch your gas stove, mister, and remember if you run your electronics 24/7, you'll see some climate change right there.

Hesitant to buy an EV because you don't trust that the grid will be able to handle all of the new demand for electricity? That AI is going to need 5 to 6 times that amount of juice but it's okay because there will probably be some awesome games for us.

These are my thoughts after reading an article from the Wall Street Journal, which I've transcribed below for convenience. The bold is mine, the rest of the article is just copied.

Article from WSJ from 12/16-17/23 entitled: The Internet Will Soon Need Way More Juice. Here's the text:

"Every company betting that artificial intelligence will transform how we work and live has a big - and growing - problem. AI is inherently ravenous for electricity.

Some experts project that global electricity consumption for AI systems could soon require adding the equivalent of a small country's worth of power generation.

Since 2010, power consumption for data centers has remained nearly flat, as a proportion of global electricity production, at about 1% of that figure, according to the International Energy Agency. But the rapid adoption of AI could represent a sea change in how much electricity is required to run the internet - specifically, the data centers that comprise the cloud, and make possible all the digital services we rely on.

This means the AI industry is poised to run the equivalent of a planetary scale experiment, according to experts both outside and inside the industry. This has some of them wringing their hands, while energy suppliers are practically salivating at the expected increase in demand.

As the WSJ reported this past week, **Microsoft is working on using its own AI to speed up the approval process for new nuclear power plants - to power that same AI.** The company is also betting on fusion power plants, inking a contract with a fusion startup that promises to deliver power within about five years - a bold move given that no one in the world has yet produced electricity from fusion.

Constellation Energy, which has **already agreed to sell Microsoft nuclear power for its data centers,** projects that AI's demand for power in the US **could be five to six times the total amount needed in the future to charge America's electric vehicles.**

Alex de Vries, a researcher at the School of Business and Economics at the Vrije Universiteit Amsterdam, projected in October that, based on current and future sales of microchips built by Nvidia specifically for AI, global power usage for AI systems could ratchet up to 15 gigawatts of continuous demand. Nvidia currently has more than 90% of the current market share for AI-specific chips in data centers, making its output a proxy for power use by the industry as a whole.

That's about the power consumption of the Netherlands, and **would require the entire output of about 15 average-size nuclear power plants.**

According to de Vries' estimates, the amount of electricity required to power the world's data centers could jump by 50% by 2027, thanks to AI alone.

Those figures might even be an underestimate, says de Vries, because his calculations didn't include the cost of cooling AI chips within data centers. Keeping these processors within the operating range is becoming increasingly difficult as they grow more powerful, and can add an additional 10% to 100% to the total amount of electricity required to run them, he adds.

Roberto Verdecchia, a computer scientist at the University of Florence who studies the power consumption of software, thinks there are additional reasons de Vries' tally of the future energy demands of AI might prove to be an underestimate.

First, companies are betting that use of AI will increase sharply. If, as they hope, it becomes a routine part of everything we do - from summarizing our every meeting to being our collaborator on every project - then demand could grow as fast as companies like Nvidia, and competitors like AMD, Intel, and Alphabet, can churn out the systems required to meet it.

Seemingly every week, there is a major new AI development. The latest was Google's new Gemini AI system, announced last week, that it claims is more powerful than any other currently on the market.

"Currently, the trend is 'bigger is better,' " says Sasha Luccioni, an AI researcher at Hugging Face, a company that makes machine-learning tools. "Every model that comes out, it's like, 'We have 10 billion more parameters than the last one."

Luccioni's research is referenced in de Vries' commentary. In her own work, Luccioni has found that there is huge variability in how much power AI systems actually use, and Luccioni isn't convinced anyone can make useful projections about whether demand for AI will continue on its current trajectory.

There is substantial debate about whether or not AI is inherently power-hungry. On one hand, AI systems must do their calculations afresh every time a user interacts with them. This is radically different than, say, the average Google search, before Google started incorporating generative AI summaries into those results.

On the other hand, there has been a great deal of research showing that AI models could be made much more efficient than they are now. Many companies are working on smaller, less power-hungry AI models that, they claim, are just as effective as the bigger ones.

All the experts I talked to agreed on this point: The main reason AI continues to require more power, even when more efficiency is possible, is that right now, the incentive for everyone in the industry is to continue building bigger, more powerful models.

As long as the competition between makes of AI continues to spur companies to use these ever more capable, ever more power-hungry models, there's no end in sight to how much more electricity the global AI industry will demand."

TLDR: AI needs a lot of electricity, wants to have us build 15 nuclear power plants JUST to make AI better.

Comments

u/jje5002 • 1 points • 2023-12-21 16:46 UTC

http://agentpalmer.com/wp-content/uploads/2014/02/SimCity-2000-Power-Plant-Options.jpg

u/una_poloma_blanca • 1 points • 2023-12-21 15:57 UTC

We should be building 100's of nuclear power plants across the country. But then the wind and solar scam artists would lose all of their subsidies and tax breaks.

u/MiserableYou6506 • 2 points • 2023-12-21 13:34 UTC

Only realisticly usable clean source of energy, atom. Known for decades

u/Eternalyskeptic • 2 points • 2023-12-21 12:33 UTC

Nuclear is clean when they want to use more power, when we want to use more power it's a toxic waste generating environmental hazard.

Essentially, shut up you useless eater.

u/[deleted] • 1 points • 2023-12-21 04:28 UTC

[removed]

u/halo_33_33 • 3 points • 2023-12-21 03:10 UTC

It'll be perfect, we can build them in canada.

u/AutoModerator • 1 points • 2023-12-21 02:43 UTC

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